

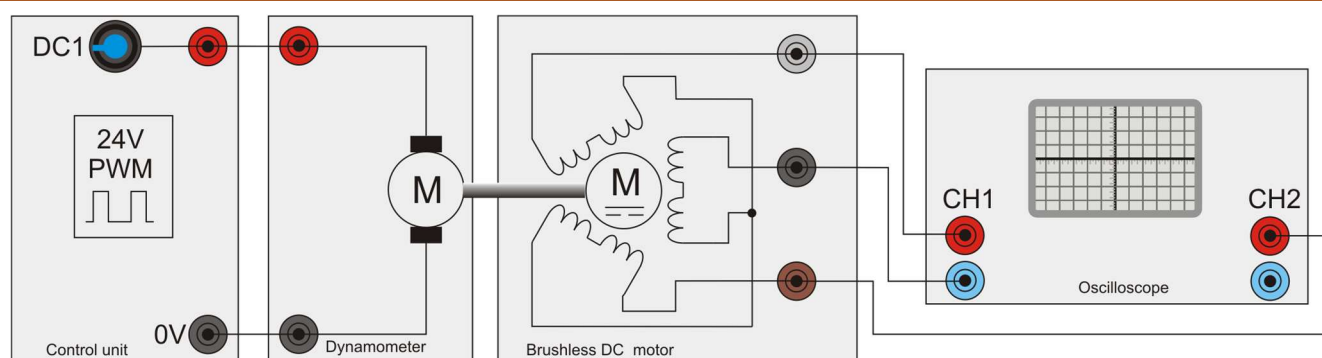
# Worksheet 21

## Three-phase generator

## Modern electrical machines



We see small wind turbines everywhere. The efficiency of three-phase over single-phase means that even small wind generators use three-phase machines to generate electricity.



**Do not use a PC-based oscilloscope.**

In this experiment, the brushless DC motor acts as a three-phase generator. It is mechanically linked to, and driven by, the dynamometer, which is supplied with a DC voltage.

### Over to you:

- 1) Set up the brushless motor as shown in the diagram above.
- 2) On your PC load 'Open Control\_DC.bat'.
- 3) Set a duty cycle of 40% on the DC1 output.
- 4) Press the 'RUN' button on the software.
- 5) Vary the DC output from 40% to 100% in 10% steps.
- 6) Make a note of the peak to peak output voltage from the brushless motor and fill in the table on the next page.
- 7) Sketch the three phase waveform on the next page.
- 8) How does the waveform relate to the motor RPM?

| 3p motor drive | Speed rpm | Vout pk-to-pk |
|----------------|-----------|---------------|
| 40%            |           |               |
| 50%            |           |               |
| 60%            |           |               |
| 70%            |           |               |
| 80%            |           |               |
| 90%            |           |               |
| 100%           |           |               |

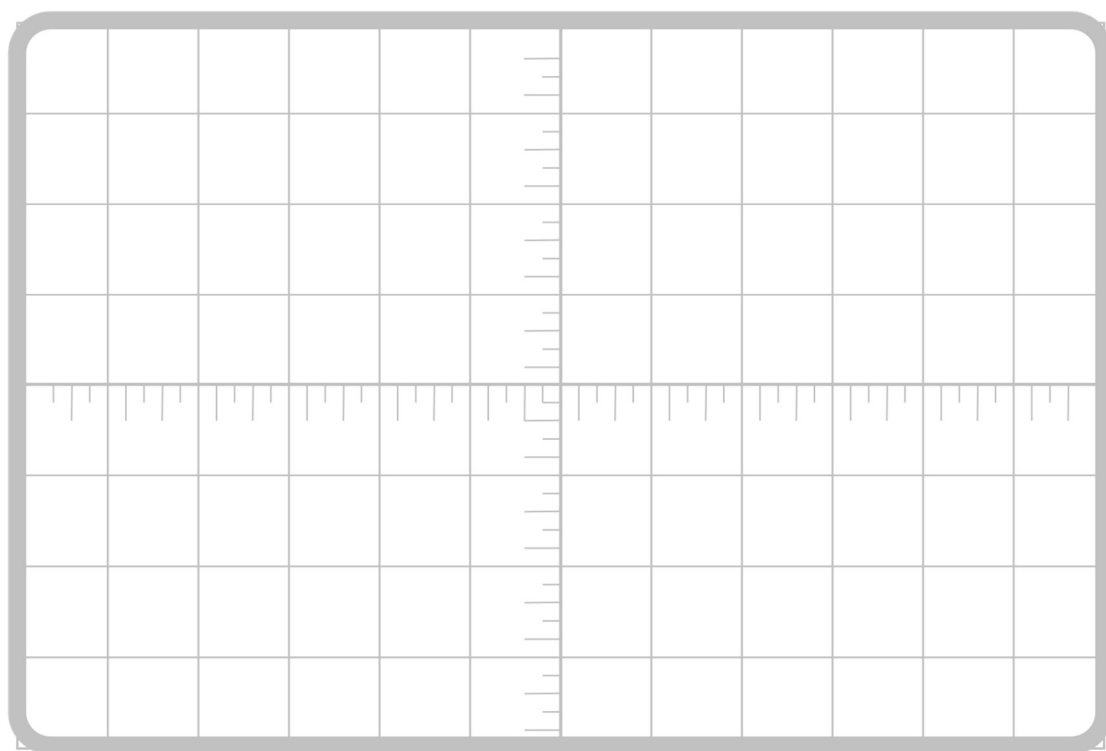
### So what?

The great thing about the brushless motor is the small size of the package and the fact that there are no brushes for ease of maintenance.

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### Over to you:

Use the tools available to investigate the efficiency of the brushless motor and compare it to that of the DC motor.